

PRABHAT DHAR

Data Architect | Cloud Data Engineer | Machine Learning Specialist

📍 London, UK ✉️ prabhatdhar32@gmail.com 💼 [linkedin.com/in/prabhat-dhar](https://www.linkedin.com/in/prabhat-dhar) 🐙 github.com/Xclipxz07

🌐 xclipxz07.github.io/Portfolio 🇬🇧 UK Work Status: Available to Work (Visa Sponsorship Required)

PROFESSIONAL SUMMARY

I started in **Pure Physics (B.Sc., University of Calcutta)** because understanding how complex systems work from first principles matters before writing a single line of code. I followed that up with a **Postgraduate Diploma in Data Science at the University of Essex**. Some people just analyze data; I build the entire pipeline from scratch. Whether it is training machine learning models that run in milliseconds, architecting automated cloud lakehouses on AWS, or building dashboards that make complex numbers obvious — I make systems faster, smarter, and completely automated. **Based in London, UK, and actively available for full-time roles (UK Skilled Worker Visa Sponsorship Required).**

EDUCATION & ACADEMIC BACKGROUND

Postgraduate Diploma in Data Science & Its Applications

University of Essex, Colchester, United Kingdom
November 2022 • 165 Credits (Distinction Level)

I.S.C (12th Grade - Pure Science & Mathematics)

St. Xavier's Institution, Kolkata, India
April 2017 • 85.0%

Bachelor of Science in Physics (Honours)

University of Calcutta, West Bengal, India
August 2021 • 7.240 CGPA

I.C.S.E (10th Grade)

St. Xavier's Institution, Kolkata, India
April 2015 • 93.8%

TECHNICAL SKILLS & ARSENAL

Programming Languages:	Python, Advanced SQL (Window Functions, CTEs, Execution Plans), R, Bash/Shell, JavaScript, LaTeX
Cloud & Lakehouse:	Amazon AWS (S3, Glue, Athena, Redshift, EMR PySpark, KMS, QuickSight), Google Cloud Platform (BigQuery)
Data Engineering:	Apache Airflow, PySpark / Spark, DuckDB, Pandas, NumPy, Snappy Parquet, Power Query, JSON/CSV/XML ETL
Machine Learning & AI:	Scikit-Learn, XGBoost, AWS AutoGluon AutoML, PyTorch, SciPy Optimization, OpenCV, OpenAI GPT-4, Gemini API
DevOps, IaC & CI/CD:	Terraform, Docker Containerization, Kubernetes, Jenkins CI/CD Automation, Git, GitHub Actions
BI & Visualization:	Microsoft Power BI, Microsoft Excel & Copilot AI, Plotly, Streamlit, Seaborn, FastAPI REST APIs

PROFESSIONAL WORK EXPERIENCE

Care Assistant & Data Analyst — EVERYCARE LTD

April 2023 – June 2026 | London, UK

- Delivered person-centred care and maintained digital care observation logs, medication updates, and activity tracking in digital care systems while ensuring strict GDPR and safeguarding compliance.
- Maintained and cleaned operational care records in Excel; engineered automated pivot models and dashboards tracking incident logs, attendance patterns, and staff allocations for shift leaders.
- Analyzed care-related trends using Power BI to visualize medication timings, risk indicators, and response times; communicated weekly operational insights to senior management.
- Wrote basic Python automation scripts to clean and preprocess raw care logs, reducing manual administrative workload by 40% and improving data accuracy for management audits.

Team Member — KFC (Demipower Ltd)

January 2021 – March 2023 | Wood Green, London, UK

- Delivered fast, friendly service in a high-pressure environment; upheld food hygiene, workplace safety, and operational quality standards.
- Handled customer transactions, managed order throughput during peak rush periods, and assisted shift leaders by maintaining inventory tracking and checklists.
- Developed cross-functional teamwork, punctuality, and time-critical communication under pressure — key transferable skills for collaborative data engineering.

FEATURED ENGINEERING & MACHINE LEARNING PROJECTS

-  **Industrial EV Digital Twin & AI Physics Surrogate** | [GitHub: Xclipxz07/ai-mechanical-engineering-digital-twin](#) Python, Scikit-Learn, SciPy, DuckDB, Parquet, Airflow, Terraform
 - Engineered a physics-informed EV digital twin ingesting multi-channel battery thermal and autonomous kinematics telemetry streams.
 - Trained AI surrogate neural regressors predicting 8-hour 3D CFD aerodynamic drag forces in **< 0.05 ms ($R^2 > 0.99$)**.
 - Implemented battery thermal gradient (ΔT) monitoring for thermal runaway prevention and applied **SciPy SLSQP optimization** for active aero flap tuning.
-  **AWS Serverless Data Analytics Lakehouse** | [GitHub: Xclipxz07/aws-serverless-data-analytics-lakehouse](#) Amazon S3, AWS Glue, Amazon Athena, Amazon QuickSight, Terraform, SQL
 - Architected an end-to-end serverless data lakehouse and SLA monitoring pipeline on AWS with automated Glue Data Catalog crawlers.
 - Slashed Athena byte-scan query costs by **85%** using Snappy Parquet partitioning and 10 GB per-query circuit breakers.
 - Executed complete 4-pillar analytical suites: Descriptive, Diagnostic, Predictive, and Prescriptive SQL for executive BI.
-  **Channel 4 Big Data Ad Revenue Pipeline** | [GitHub: Xclipxz07/channel4-bigdata-ad-pipeline](#) PySpark, AWS EMR, Apache Airflow, DuckDB, Snappy Parquet, Terraform
 - Automated nightly 2:00 AM Big Data lakehouse ingesting multi-format ad logs (JSON, CSV, XML) across Smart TVs and Web players.
 - Orchestrated PySpark transformations on AWS EMR with Apache Airflow DAGs, computing regional eCPM yield curves in sub-second queries.
-  **Sales Forecasting System for DevOps, MLOps & GenAI** | [GitHub: Xclipxz07/DATA-ANALYSIS-FOR-DEVOPS-MLOPS-GENAI](#) FastAPI, OpenAI GPT-4, Scikit-learn, Docker, Kubernetes, Jenkins, Power BI
 - Built production-ready sales forecasting system combining ML, generative AI (OpenAI GPT-4), and full-lifecycle DevOps automation.
 - Containerized REST API with Docker, orchestrated on Kubernetes with horizontal scaling, and automated testing via Jenkins CI/CD.
-  **Australian Rainfall Forecasting with PyCaret & ML (Dissertation)** | [GitHub: Xclipxz07/Rainfall-prediction](#) Python, PyCaret AutoML, Scikit-learn, XGBoost, Pandas, Statsmodels, LaTeX
 - Conducted academic research dissertation forecasting weather patterns across Australia; engineered automated EDA and outlier handling.
 - Trained and evaluated XGBoost, Random Forests, and Gradient Boosting models using PyCaret AutoML with complete ROC-AUC diagnostics.
-  **NHS Patient Healthcare Analytics Lakehouse** | [GitHub: Xclipxz07/nhs-patient-healthcare-analytics-lakehouse](#) Terraform, AWS S3, AWS Glue, Amazon Athena, Apache Airflow, Snappy Parquet
 - Developed an automated healthcare lakehouse on AWS using Terraform and Airflow for hospital emergency A&E waiting-time analytics.
 - Enforced strict GDPR / HIPAA compliance by implementing SHA-256 cryptographic hashing to pseudonymize sensitive patient identifiers.

VERIFIED CERTIFICATIONS & CLEARANCES

- English SELT ISE 1:** Trinity College London ([Distinction in Speaking, Listening & Writing](#))
 - Microsoft Data Analysis:** [Verify \(ID: VXYOBGDSIQOO\)](#)
 - Google AI Professional:** [Verify \(ID: FW6A17MGN8U3\)](#)
 - AWS Data Analytics:** [Verify \(ID: 0R8IQQB4CTI1\)](#)
 - Image Classification AutoGluon:** [Verify \(ID: MVVF8ARA6H2C\)](#)
- Enhanced DBS Certification:** Active & Verified ([All UK background checks cleared](#))
 - Gemini for Developers:** [Verify \(ID: DMVKPELEY3J2\)](#)
 - AI for Mechanical Engineers:** [Verify \(ID: BUUPCEOS9DG1\)](#)
 - Scikit-Learn ML Classification:** [Verify \(ID: 8KLLIRRLXTWN\)](#)
 - Data Engineering Guides:** [8 Master Blueprints](#)

 English: Fluent / Professional (Trinity College London SELT ISE 1 — Distinction)	 Hindi: Native Proficiency	 Bengali: Native Proficiency
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